

Forward-difference Formula

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} - \frac{h}{2} f''(\xi)$$

where ξ between x_0 and $x_0 + h$



Backward-difference Formula

$$f'(x_0) = \frac{f(x_0) - f(x_0 - h)}{h} + \frac{h}{2} f''(\xi)$$

where ξ between x_0 and $x_0 - h$



Two-Point Formula

Example

Use table 4.2 and **forward-difference** formula to approximate $f'(2)$ with $h = 0.1$

$$h = 0.1 \quad f'(2) \approx \frac{1}{(0.1)} \{f(2.1) - f(2)\}$$

Use table 4.2 and **backward-difference** formula to approximate $f'(2)$ with $h = 0.2$

$$h = 0.2 \quad f'(2) \approx \frac{1}{(0.2)} \{f(2) - f(1.8)\}$$

Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

ileri
fark

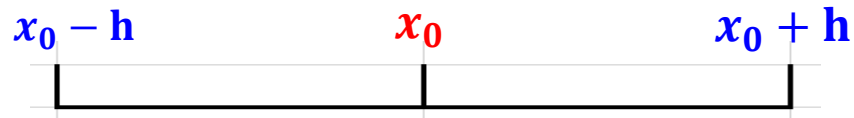
geri
fark

Üç Nokta Formülleri

Three-Point Midpoint Formula

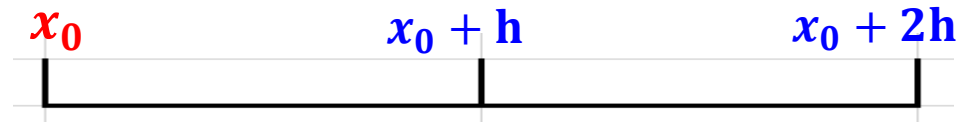
$$f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6}f^{(3)}(\xi_1),$$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

**Three-Point Endpoint Formula**

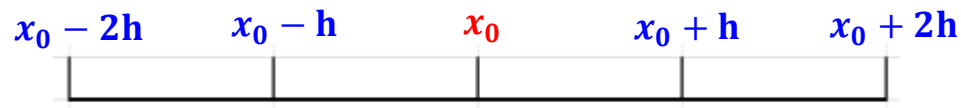
$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0),$$

where ξ_0 lies between x_0 and $x_0 + 2h$.

**Five-Point Midpoint Formula**

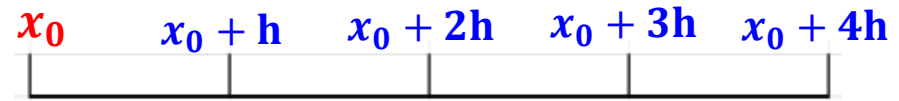
$$f'(x_0) = \frac{1}{12h}[f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)] + \frac{h^4}{30}f^{(5)}(\xi)$$

where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.

**Five-Point Endpoint Formula**

$$f'(x_0) = \frac{1}{12h}[-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - f(x_0 + 4h)] + \frac{h^4}{5}f^{(5)}(\xi),$$

where ξ lies between x_0 and $x_0 + 4h$.



Three-Point Midpoint Formula

$$f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6}f^{(3)}(\xi_1),$$

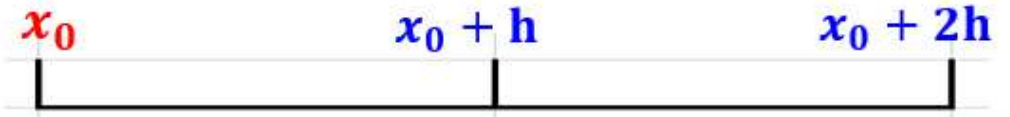
where ξ_1 lies between $x_0 - h$ and $x_0 + h$.



Three-Point Endpoint Formula

$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0),$$

where ξ_0 lies between x_0 and $x_0 + 2h$.



Example

Values for $f(x) = xe^x$ are given in Table 4.2.

Use table 4.2 and three-point **endpoint** formula to approximate $f'(2)$ with $h = 0.1$

$$\begin{aligned} f'(2) &\approx \frac{1}{0.2}[-3f(2.0) + 4f(2.1) - f(2.2)] : \\ &= 5[-3(14.778112) + 4(17.148957) - 19.855030] \\ &= 22.032310 \end{aligned}$$

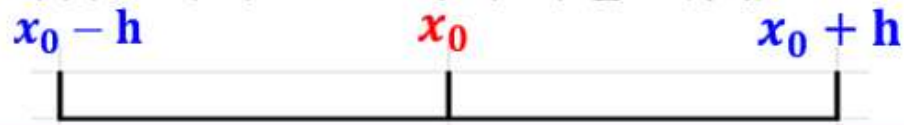
Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Three-Point Midpoint Formula

$$f'(x_0) = \frac{1}{2h}[f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6}f^{(3)}(\xi_1),$$

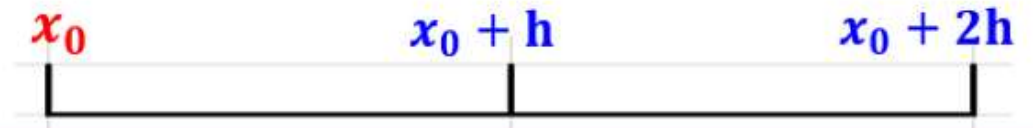
where ξ_1 lies between $x_0 - h$ and $x_0 + h$.



Three-Point Endpoint Formula

$$f'(x_0) = \frac{1}{2h}[-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3}f^{(3)}(\xi_0),$$

where ξ_0 lies between x_0 and $x_0 + 2h$.



Example

Use table 4.2 and three-point **midpoint** formula to approximate $f'(2)$ (consider all possible cases)

$$h = 0.1 \quad f'(2) \approx \frac{1}{2(0.1)}\{f(2.1) - f(1.9)\}$$

$$h = 0.2 \quad f'(2) \approx \frac{1}{2(0.2)}\{f(2.2) - f(1.8)\}$$

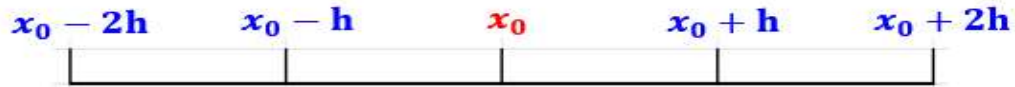
Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Five-Point Midpoint Formula

$$f'(x_0) = \frac{1}{12h} [f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)] + \frac{h^4}{30} f^{(5)}(\xi)$$

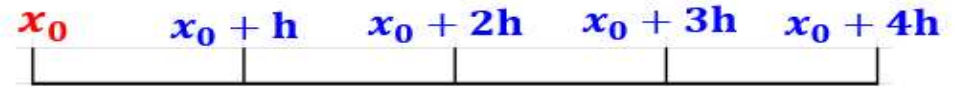
where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.



Five-Point Endpoint Formula

$$f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - 3f(x_0 + 4h)] + \frac{h^4}{5} f^{(5)}(\xi)$$

where ξ lies between x_0 and $x_0 + 4h$.



Example

Use table 4.2 and five-point **midpoint** formula to approximate $f'(2)$ with $h = 0.1$

$$\begin{aligned} f'(2) &\approx \frac{1}{1.2} [f(1.8) - 8f(1.9) + 8f(2.1) - f(2.2)] \\ &= \frac{1}{1.2} [10.889365 - 8(12.703199) + 8(17.148957) - 19.855030] \\ &= \\ &22.166999 \end{aligned}$$

Table 4.2

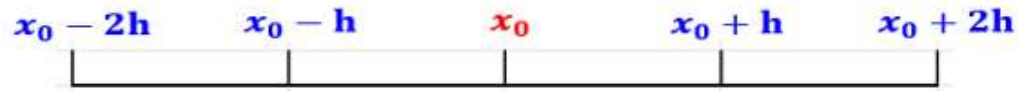
x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

The only five-point formula for which the table gives sufficient data is the midpoint formula

Five-Point Midpoint Formula

$$f'(x_0) = \frac{1}{12h} [f(x_0 - 2h) - 8f(x_0 - h) + 8f(x_0 + h) - f(x_0 + 2h)] + \frac{h^4}{30} f^{(5)}(\xi)$$

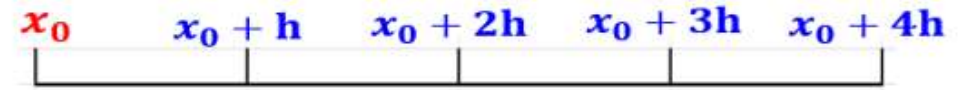
where ξ lies between $x_0 - 2h$ and $x_0 + 2h$.



Five-Point Endpoint Formula

$$f'(x_0) = \frac{1}{12h} [-25f(x_0) + 48f(x_0 + h) - 36f(x_0 + 2h) + 16f(x_0 + 3h) - 3f(x_0 + 4h)] + \frac{h^4}{5} f^{(5)}(\xi)$$

where ξ lies between x_0 and $x_0 + 4h$.



Example

Compute the actual error

The data given in Table 4.2 are taken from the function $f(x) = xe^x$

The actual error = $|22.166999 - f'(2)| = |22.166999 - 22.167168| = 1,69 \cdot 10^{-4}$

multiple hat a

Comparison Study (Karsilasturma çalismasi)

Example

	$f'(2.0) \approx$	True Error
The true value	22.167168	
Three-point endpoint with $h = 0.1$	22.03231	1.35×10^{-1}
Three-point endpoint with $h = -0.1$	22.054525	1.13×10^{-1}
Three-point midpoint with $h = 0.1$	22.22879	-6.16×10^{-2}
Three-point midpoint with $h = 0.2$	22.414163	-2.47×10^{-1}
Five-point midpoint with $h = 0.1$	22.166999	1.69×10^{-4}

Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Table 4.2 represents values of $f(x) = xe^x$

If we had no other information we would accept the five-point midpoint approximation using $h = 0.1$ as the most accurate, and expect the true value to be between that approximation and the three-point mid-point approximation that is in the interval $[22.166, 22.229]$.

Bu çalışmada,

1) Üs nokta us formülünde $h=0.1$ ve $h=-0.1$

olarak $f'(2)$ hesaplanıyor. Sonrasında mutlak hata (true error) hesaplanıyor.

2) Üs nokta orta formülünde $h=0.1$ ve $h=0.2$

olarak $f'(2)$ hesaplanıyor. Benzer şekilde mutlak hata hesaplanıyor.

3) Beş nokta formülünde $h=0.1$ alınarak $f'(2)$ hesaplanıyor.

* Yapılan bu çalışma, $h=0.1$ olarak elde edilen sonucu gerçek değere en yakın değer olduğunu gösteriyor.

* Ayrıca, bu çalışmayla $f'(2)$ 'nin gerçek değerini bilmeden bile

$[22.166, 22.229]$ aralığında olduğunu tahmin edebiliriz.

Second Derivative Midpoint Formula

$$f''(x_0) = \frac{1}{h^2}[f(x_0 - h) - 2f(x_0) + f(x_0 + h)] - \frac{h^2}{12}f^{(4)}(\xi),$$

where $x_0 - h < \xi < x_0 + h$.

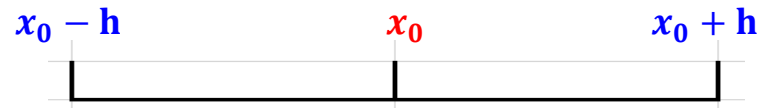


Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

$(e^x + xe^x)' = e^x + e^x + xe^x = e^x(x+2)$

Example

Use table 4.2 and Second Derivative Midpoint Formula to approximate $f''(2)$ with $h = 0.1, 0.2$

$$f''(2) \approx \frac{1}{0.01}[f(1.9) - 2f(2.0) + f(2.1)] :$$

$h = 0.1$

$$= 100[12.703199 - 2(14.778112) + 17.148957]$$

$$= 29.593200$$

$$f''(2) \approx \frac{1}{0.04}[f(1.8) - 2f(2.0) + f(2.2)]$$

$$= 25[10.889365 - 2(14.778112) + 19.855030]$$

=

$$\text{True Error } 29.704275$$

The true value $4e^2 \rightarrow$	$f''(2.0) \approx$	
	29.556224	
midpoint with $h = 0.1$	29.5932	$-3.70 \text{ e-}2$
midpoint with $h = 0.2$	29.704275	$-1.48 \text{ e-}1$